Gramática:

0 $S' \to Program\ \$$
1 $Program \to Program\ Stmt$
2 $Program \to Stmt$
3 $Stmt \to id = Expr\ ;$
4 $Expr \to Expr + Term$
5 $Expr \to Term$
6 $Term \to Term * Factor$
7 $Term \to Factor$
8 $Factor \to num$
9 $Factor \to id$

$I_0 = Closure([S' \to \bullet Program\ \$])$

$= \{ [S' \to \bullet Program, \$],$
$[Program \to \bullet Program\ Stmt, id/\$],$
$[Program \to \bullet Stmt, id/\$],$
$[Stmt \to \bullet id = Expr\ ;, id/\$] \}$

$I_1 = goto(I_0, id)$

$= \{ [Stmt \to id \bullet = Expr\ ;, id/\$] \}$

$I_2 = goto(I_0, Program)$

$= \{ [S' \to Program \bullet, \$],$
$[Program \to Program \bullet Stmt, id/\$],$
$[Stmt \to \bullet id = Expr, id/\$] \}$

$I_3 = goto(I_0, Stmt)$

$= \{ [Program \to Stmt \bullet, id/\$] \}$

$I_4 = goto(I_1, =)$

$= \{ [Stmt \to id = \bullet Expr, id/\$],$
$[Expr \to \bullet Expr + Term, ;/+],$
$[Expr \to \bullet Term, ;/+],$
$[Term \to \bullet Term * Factor, ;/+/*],$
$[Term \to \bullet Factor, ;/+/*],$
$[Factor \to \bullet num, ;/+/*],$
$[Factor \to \bullet id, ;/+/*] \}$

$I_5 = goto(I_2, Stmt)$

$= \{ [Program \to Program\ Stmt \bullet, id/\$] \}$

$I_6 = goto(I_4, Factor)$

$= \{ [Factor \to id \bullet, ;/+/*] \}$

$I_7 = goto(I_4, num)$

$= \{ [Factor \to num \bullet, ;/+/*] \}$

$I_8 = goto(I_4, Expr)$

$= \{[Stmt \to id = Expr \bullet ;, id/\$],$
$[Expr \to Expr \bullet + Term, ;/*]\}$

$I_9 = goto(I_4, Term)$

$= \{[Expr \to Term \bullet, ;/*],$
$[Term \to Term \bullet * Factor, ;/+/*]\}$

$I_{10} = goto(I_4, Factor)$

$= \{[Term \to Factor \bullet, ;/+/*]\}$

$I_{11} = goto(I_8, ;)$

$= \{[Stmt \to id = Expr; \bullet, id/\$]\}$

$I_{12} = goto(I_8, +)$

$= \{[Expr + \bullet Term, ;/+],$
$[Term \to \bullet Term * Factor, ;/+/*],$
$[Term \to \bullet Factor, ;/+/*],$ REPETIDO
$[Factor \to \bullet num, ;/+/*],$ REPETIDO
$[Factor \to \bullet id, ;/+/*]\}$ REPETIDO

$I_{13} = goto(I_9, *)$

$= \{[Term \to Term * \bullet Factor, ;/+/*],$
$[Factor \to \bullet num, ;/+/*],$
$[Factor \to \bullet id, ;/+/*]\}$

$I_{14} = goto(I_{12}, Term)$

$= \{[Expr \to Expr + Term \bullet, ;/+],$
$[Term \to Term \bullet * Factor, ;/+/*]\}$
REPETIDO

$I_{15} = goto(I_{13}, Factor)$

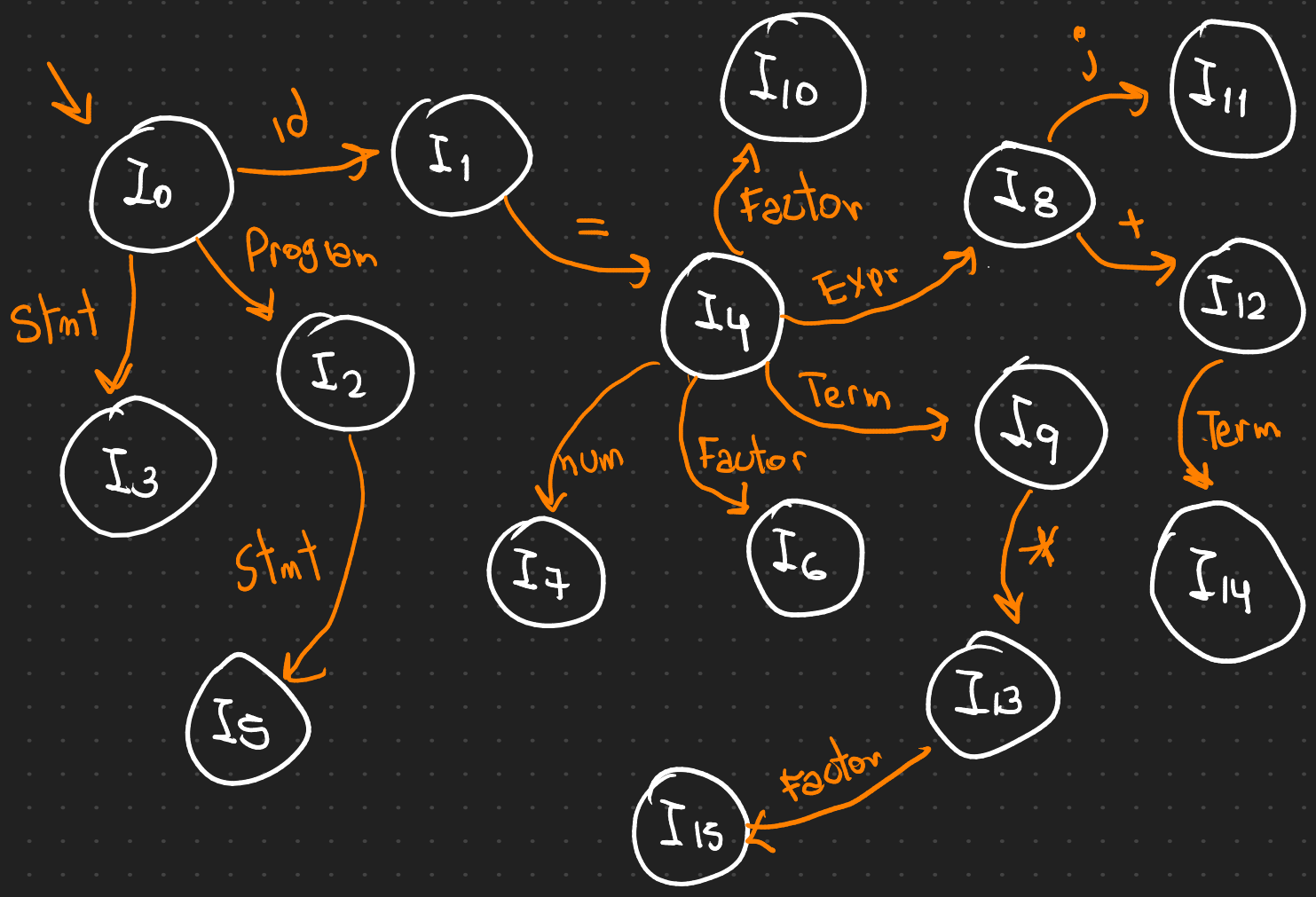
$= \{[Term \to Term * Factor \bullet, ;/+/*]\}$

Total: 16 estados

Conversión de LR(1) a LALR(1): Todas las kernels (núcleos) son distintos

$\to$ LR(1) = LALR(1) para esta gramática

AUTÓMATA


I0
id
I1
=
I4
Factor
I10
Program
I2
Stmt
I3
Stmt
I5
num
I7
Factor
I6
Expr
I8
Term
I9
;
I11
+
I12
Term
I14
*
I13
factor
I15

ACTION y GOTO Table

	id	=	;	+	*	num	$	Prog	Stmt	Expr	Term	Factor
10	S1							2	3			
11		S4						—	—	—	—	—
12	S1						acc	—	5			
13	R2						R2	—	—			
14	S2					S7		—	—	8	9	10
15	R1						R1					
16	—	—	R9	R9	R9							
17	—	—	R8	R8	R8							
18	—	—	S11	S12								
19	—	—	R5	R5	S13							
110	—	—	R7	R7	R7							
111	R3	—	—	—	—	—	R3					
112	S6	—	—	—	—	S7	—	—	—	—	14	10
113	S6	—	—	—	—	S7	—	—	—	—	—	15
114			R4	R4	S1							
115			R6	R6	R6							